MIL-C-5191B
AMENDMENT 4
4 August 1986
SUPERSEDING
AMENDMENT 3
12 November 1980

MILITARY SPECIFICATION

COMPUTER AND TRANSMITTER, TRU AIRSPEED,
PITOT STATIC, TYPE A-2

This amendment forms a part of MIL-C-5191B, dated 19 Mar 52, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Delete "2.1" and add:
"2.1 Government documents."

Add:
"2.1.1 Specifications, standards, and handbooks. The following specifications, standard, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

Under SPECIFICATIONS, Federal, delete "QQ-P-416" and add the following:
"PPP-B-636 Boxes, Shipping, Fiberboard"

* Under SPECIFICATIONS, Military, delete "MIL-D-5028", "MIL-E-5272", "MIL-P-5633" and "MIL-M-7911".

* Add:
"DOD-D-1000 Drawings, Engineering And Associated List
MIL-P-116 Preservation-Packaging, Methods Of
MIL-A-8625 Anodic Coatings, For Aluminum and Aluminun Alloys
MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited"

* "Under Air Force-Navy Aeronautical, delete, "AN-QQ-A-696."

Add boxed paragraph to center bottom of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

AMSC N/A FSC 6610

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* Under STANDARDS, Military, add:

"MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standard And Specifications, Order Of Precedence For The Selection Of
MIL-STD-726	Packaging Requirements Codes
MIL-STD-794	Parts And Equipment, Procedures For Packaging and Packaging Of
MIL-STD-810C	Environmental Test Methods
MS3102	Connector, Recepticle, Electric, Box Mounting, Solder, and Type
MS24693	Srew, Machine, Flat Countersunk Head, 100 Deg, Cross Recessed, UNC-2A & UNF-2A
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument"

*Under Drawings, delete "AN507", "AN3102" and "AND10401"

Under PUBLICATIONS, Air Force - Navy Aeronautical Bulletin, delete "No. 143".

* Paragraph 3.1, delete entirely and add:

"3.1 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification."

* Paragraph 3.1.4. delete entirely.

PAGE 3

* 3.2 Line 2 delete "AND10401" and add: "MS33585".

3.2.2.1: In the last sentence, delete "150 knots (true) or above 600 knots" and substitute "150 knots or above 650 knots true".

PAGE 4

FIGURE 1. Case - computer and transmitter: Delete and substitute new FIGURE 1, which follows.

* Figure 1, delete "Electrical Receptacle AN 3102A-145-5P" and add: "Electrical Receptacle MS 3102A-145-5P".

PAGE 6

* Paragraph 3.3, delete "MIL-D-5028" and add: "DOD-D-1000".

* 3.7.1 Line 2 delete "AN-QQ-A-696" and add: "MIL-A-8625".

PAGE 7

3.7.2: Delete existing paragraph and substitute the following new paragraph:

"3.7.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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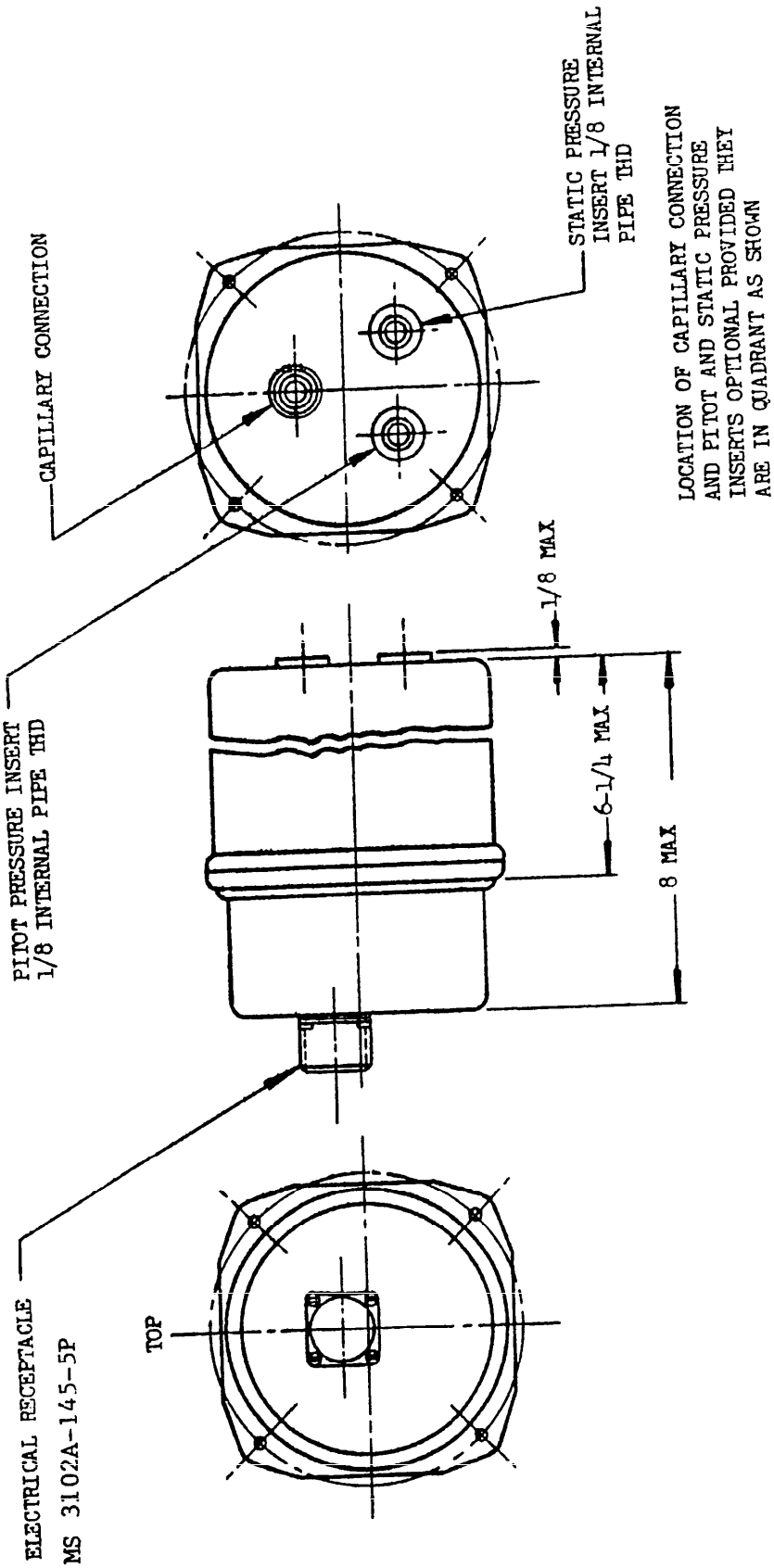


FIGURE 1. Case - computer and transmitter

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3.8 and 3.8.1: Delete these paragraphs in their entirety and substitute the following:

"3.8 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130."

PAGE 10

* Paragraph 4.3.3.1, delete "MIL-E-5272" and add: "MIL-STD-810C".

* Paragraph 4.3.3.1.1, delete "Procedure II" and add: "Method 502.1, Procedure I".

* Paragraph 4.3.3.1.2, delete "Procedure I" and add: "Method 501.1, Procedure I".

* Paragraph 4.3.3.1.3, delete "Procedure II" and add: Method 500.1, Procedure I".

* Paragraph 4.3.3.1.4, delete entirely except table and add: "4.3.3.1.4 Vibration. Vibration: Method 514.2, Procedure I, Part 1, Figure 514.2-2, Curve D. The test shall be conducted while a differential pressure of 400 knots is applied to the computer and transmitter. The difference between the indication of the pointer at the beginning and end of the three hour vibration period shall not exceed the tolerance specified in Table I, and no looseness in the mechanism nor damage shall result in any part of the computer and transmitter".

* Paragraph 4.3.3.1.5, delete entirely and replace with: "4.3.3.1.5 Humidity. Humidity: Mehtod 507.1, Procedure I, except the internal temperature raised to 71°C instead of 65°C and relative humidity throughout the entire cycle shall be 95 percent. There shall be no evidence of corrosion or rusting of the interal or external parts of the computer and transmitter".

* Paragraph 4.3.3.1.6, delete, "Procedure I" and add: "Method 508.2".

PAGE 11

4.4.2.3.1: Delete the last sentence and add:
"The indicated airspeed at the minimum voltage shall be 238 knots; the tolerance specified in table IV shall apply."

PAGE 13

TABLE II: Add the following test points to this table:

"TABLE II

Test temperature °C	Pressure altitude feet	Indicated airspeed knots	True airspeed knots	Indication degrees	Tolerance degrees
±40	0	650*	630	191	5
	5,000	600*	627	190	5

"

* Test Points

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PAGE 14

TABLE III: Add the following test points to this table:

±20	0	650*	609	183	5
	5,000	600*	607	182	5

"

* Test Points

PAGE 15

TABLE IV: Add the following test points to this table:

"TABLE IV

0	5,000	650*	627	190	5

"

*Test Points

In the "10,000 Pressure altitude feet" section, in the columns headed "True airspeed knots" and "Indication degrees" delete the entries "211 through 410" and "40 through 112" respectively and substitute the following:

"True airspeed knots	Indication degrees
231	47
287	67
340	86
393	105
443	123

PAGE 16

TABLE V: In the "True airspeed knots" column, delete "443" and substitute "448". Add the following test points to this table:

"TABLE V

-20	10,000	600*	603	181	5
	20,000	500*	599	179	6

"

* Test Points

PAGE 18

TABLE VII: In the column headed "Test temperature ^{o}C" add "60".

PAGE 19

4.4.2.4.4.1: Delete the equation for "V_{true}" and substitute the following:

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$$"V_{true} = 87.0114 \sqrt{T_o \left[\left(1 + \frac{dp}{P}\right)^{0.2857} - 1\right]}" \quad (1)$$

PAGE 20

4.4.3.5 Seasoning: In the last sentence delete "table V" and substitute "table III".

PAGE 21

Add the following new paragraphs to Section 4:

"4.6 Packaging inspection

4.6.1 Quality conformance. The inspection of the preservation-packaging and exterior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.6.2 First article. When the unit container is capable of serving as the shipping container, First Article Inspection, and rough handling tests as outline in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction."

Delete Section 5 in its entirety and substitute the following new section:

"5 PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified by contracting activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.2 of this specification.

5.1.2 level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

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5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601 plywood box, shall be used for the shipping container.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather-resistant.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the suppply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129. Nomenclature shall be: Computer and Transmitter, True Airspeed Pitot Static, Type A-2."

PAGE 23

6.1 Intended use: Delete "150 to 600 knots" and add: "150 to 650 knots".

Add new paragraph 6.4:
"6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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